

HEAT TRANSFER THROUGH COMPOSITE WALL

1 Objective

The objective of this simulation is to analyze steady-state heat conduction through a two-layer composite wall using ANSYS Fluent 2019 and determine:

- Temperature distribution
- Average temperature at the interface (Wall A)
- Average temperature at the outer surface (Wall B)

The numerical results are compared with analytical calculations.

2 Problem Description

A two-layer wall is subjected to:

- Left surface temperature = 200°C
- Right surface convection:
 - Ambient temperature = 30°C
 - Heat transfer coefficient (h) = 40 W/m²K

Material properties:

- Layer 1:
 - Thickness = 0.05 m
 - Thermal conductivity = 0.08 W/mK
- Layer 2:
 - Thickness = 0.15 m
 - Thermal conductivity = 0.074 W/mK

Height of wall = 0.10 m

Heat transfer is assumed steady and two-dimensional.

3 Geometry Creation (DesignModeler)

The geometry was created in ANSYS DesignModeler.

- Units: Meters
- A rectangular wall of size 0.20 m × 0.10 m was created.
- The wall was divided into two parts at $x = 0.05$ m to represent two materials.

- Two surface bodies were generated.

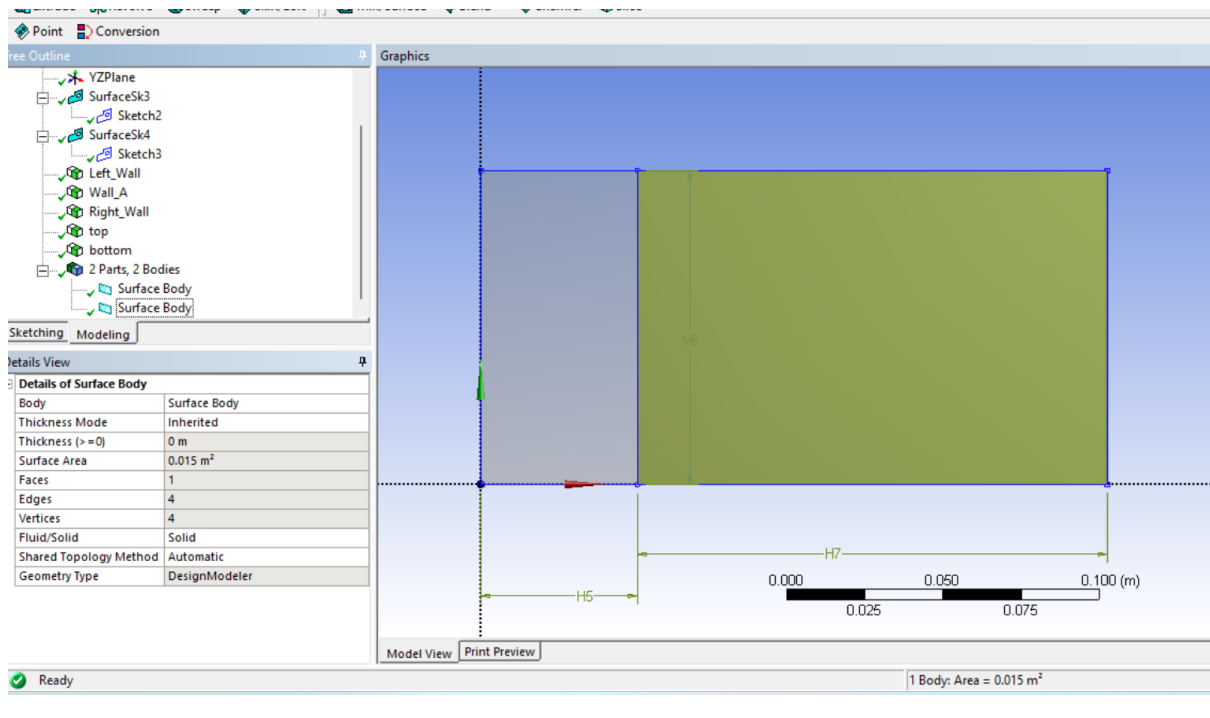


Figure 1 Geometry

4 Mesh Generation

Meshing was performed in ANSYS Meshing.

- Physics Preference: CFD
- Element size: 0.005 m
- Quadrilateral dominant mesh
- Two separate bodies confirmed

Total elements: __1970__

Total nodes: __20096__

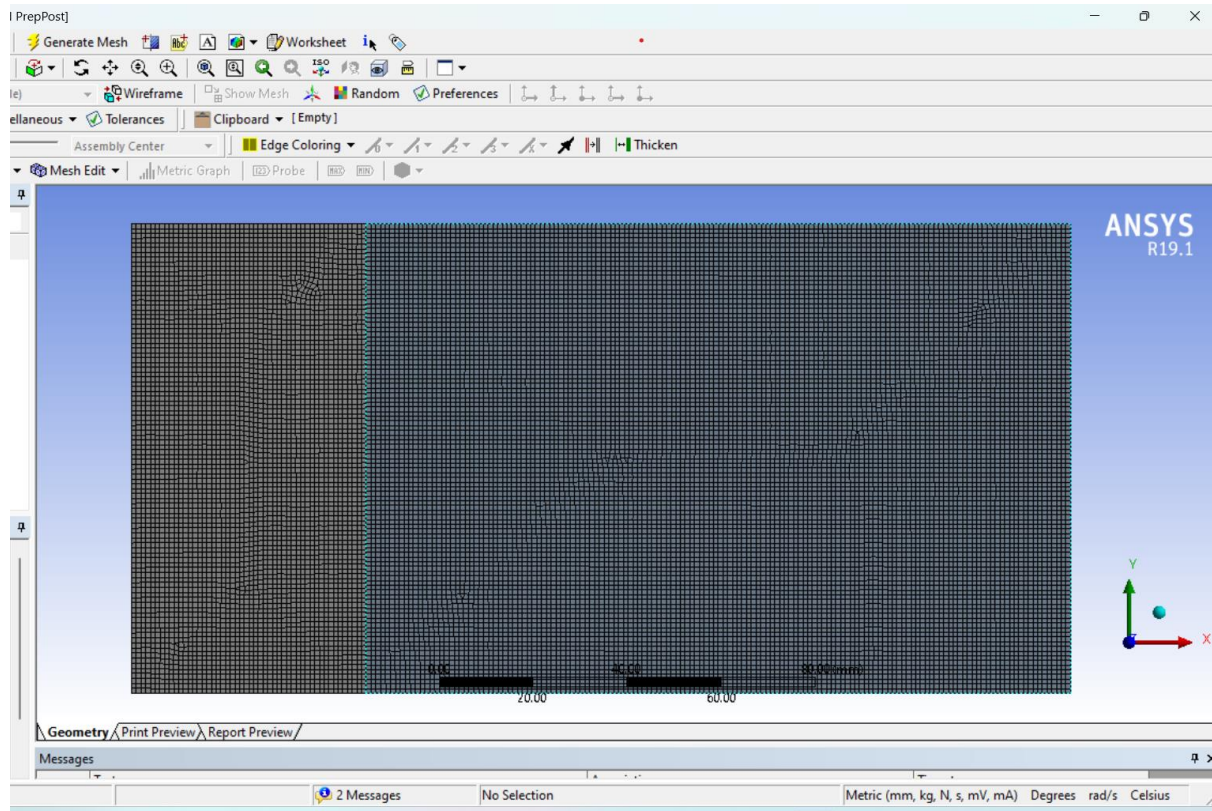


Figure 2 Mesh

5□ Fluent Setup

5.1 General Settings

- Solver type: Pressure-Based
- Time: Steady
- 2D model
- Energy Equation: Enabled

5.2 Materials

Two solid materials were defined:

Material 1:

- Thermal conductivity = 0.08 W/mK

Material 2:

- Thermal conductivity = 0.074 W/mK

Each material was assigned to its respective solid zone.

5.3 Boundary Conditions

Left Wall:

- Type: Wall
- Temperature = 200°C

Right Wall:

- Type: Wall
- Thermal condition: Convection
- $h = 40 \text{ W/m}^2\text{K}$
- Ambient temperature = 30°C

Top and Bottom:

- Heat Flux = 0 (Adiabatic)

Interface Wall:

- Default coupled condition
-

6□ Solution Method

- Pressure-Velocity Coupling: SIMPLE
- Spatial Discretization (Energy): Second Order Upwind
- Initialization: Hybrid Initialization
- Iterations: 500

Residuals converged below 10^{-6} .

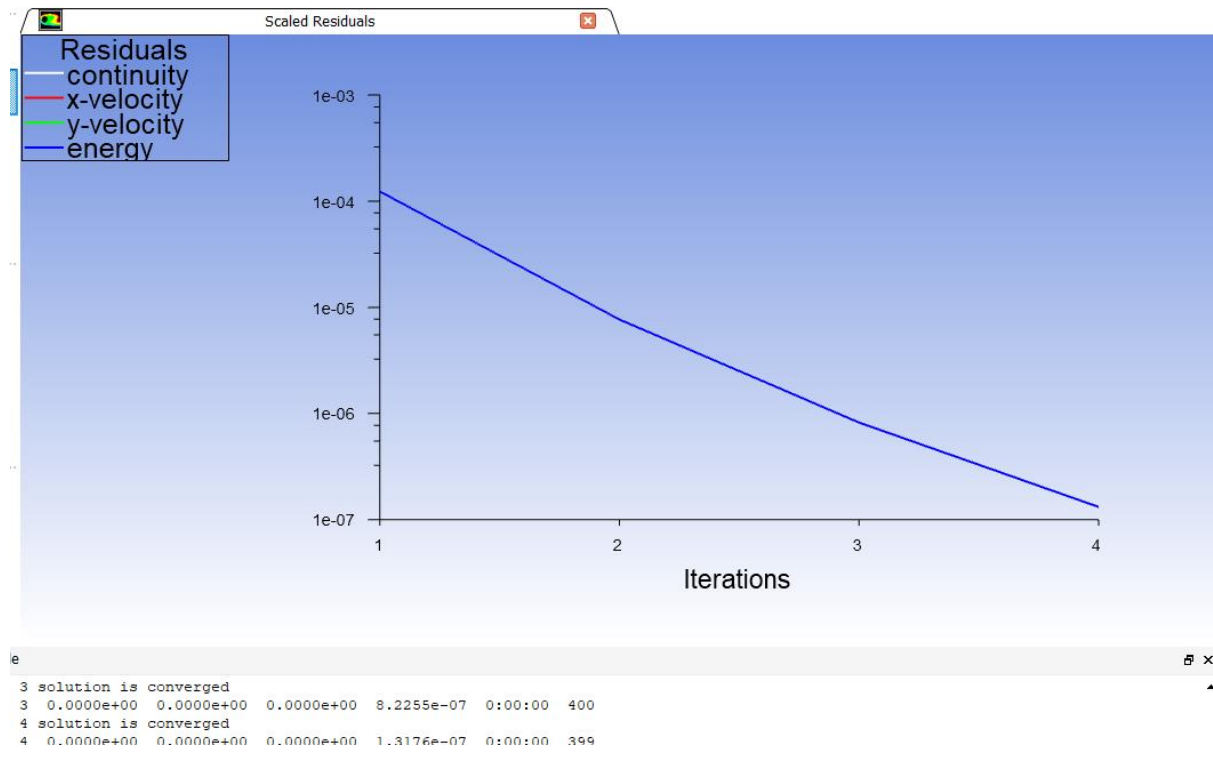


Figure 3 Residual Graph

7□ Results and Discussion

7.1 Temperature Contour

The temperature contour shows a smooth decrease in temperature from left (200°C) to right side.

Different temperature gradients are observed in both layers due to different thermal conductivities.

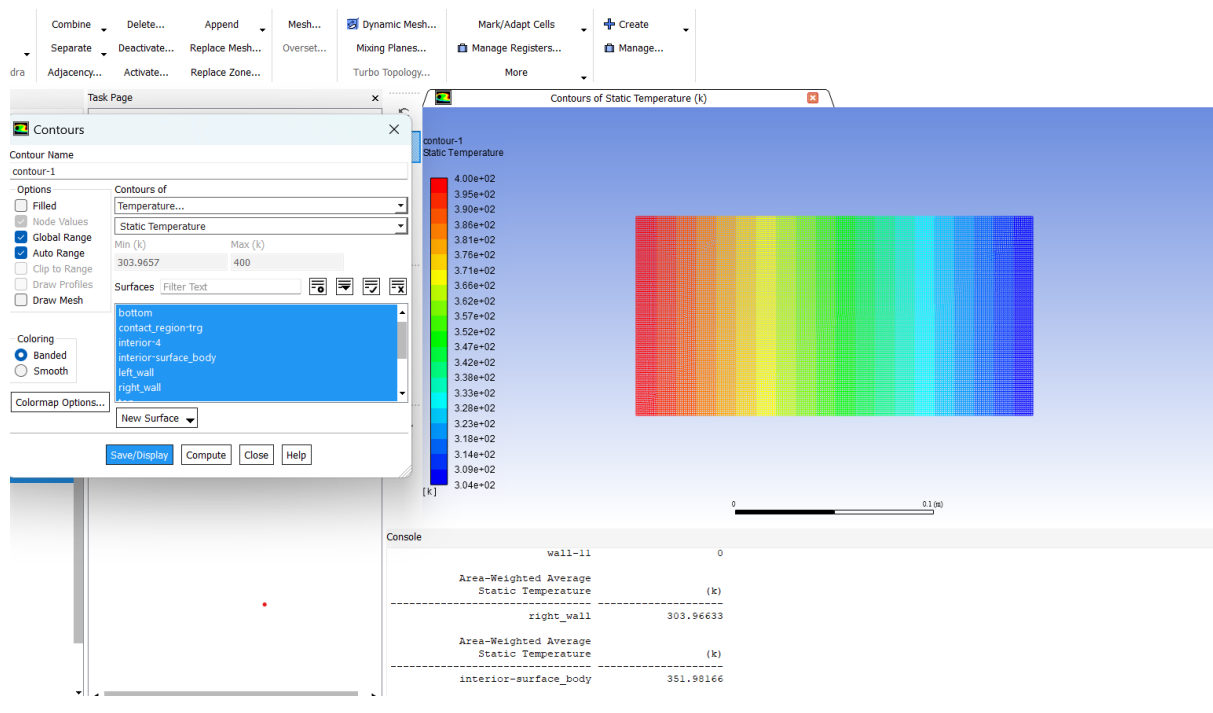


Figure 4 Result

7.2 Interface Temperature (Wall A)

Area-weighted average temperature at interface:

$$T_A = 160.9 \text{ } ^\circ\text{C}$$

7.3 Outer Surface Temperature (Wall B)

Area-weighted average temperature at right wall:

$$T_B = 31.9 \text{ } ^\circ\text{C}$$

9 Comparison

Parameter	Analytical	ANSYS
Interface Temp	160.31	160.9
Outer Temp	30	31.9

The numerical results show good agreement with the analytical solution.

10 Conclusion

The simulation successfully modeled steady heat conduction through a composite wall. The temperature decreases linearly in each layer, with slope variation due to different thermal conductivities.

The ANSYS results closely match analytical calculations, validating the numerical model.